



Course: Signals & Systems  
 Program: BS(Electrical Engineering)  
 Duration: 60 Minutes  
 Paper Date: 8th April, 2017  
 Section: ALL  
 Exam: Midterm-II

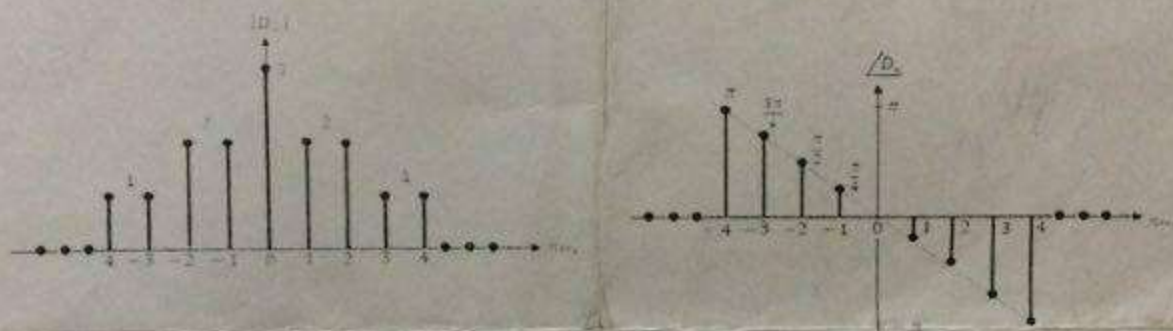
Course Code: EE223  
 Semester: Spring 2017  
 Total Marks: 40  
 Weight: 15%  
 Page(s): 1  
 Roll No: 215-4522

Instruction/Notes: Carefully read the paper and attempt each question only once. Attempt all parts of a question together.

Question #1: CLO (3)

15 Marks

Exponential Fourier Spectra of a periodic signal  $x(t)$  are given below:



- Find the exponential Fourier series using the above spectra.
- Find the compact trigonometric Fourier series of  $x(t)$ .
- Sketch the spectra of compact trigonometric Fourier series using part (b).
- Find the bandwidth of the signal.

10 Marks

Question #2 CLO (3)

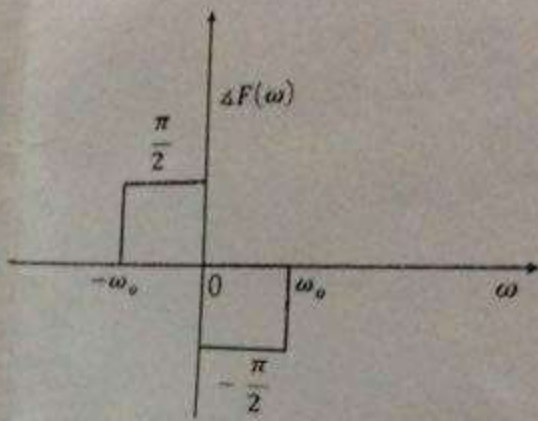
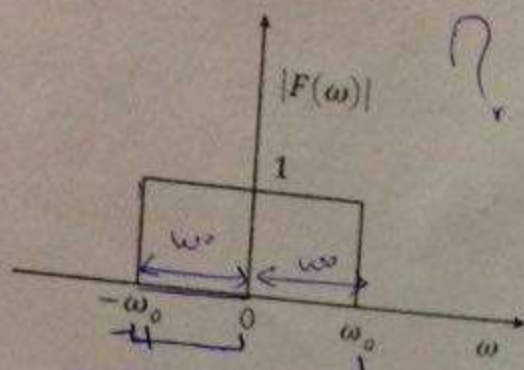
For the following continuous time periodic signals:

- $x(t) = \sin 50\pi t + \cos 30t$
  - $y(t) = 3 + 2\cos\left(\frac{5\pi}{2}t\right) + \sin\left(\frac{6\pi}{5}t + \frac{\pi}{3}\right) + 7\cos\left(\frac{\pi}{4}t + \frac{\pi}{4}\right)$   $T=8?$
- Find the fundamental frequency and time period.  $1 \text{ } T=2\pi?$
  - Identify which harmonics are present in the signal.  $\pi/4 \text{ } T=8?$



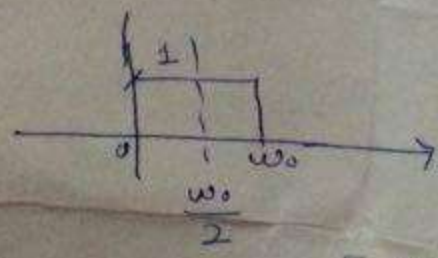
Question #3 CLO (4)

Find and simplify the Inverse Fourier transform  $f(t)$  from the magnitude and phase spectra given below



width =  $2\omega_0$

$\omega_0$



$$\left[ \frac{\omega - \frac{\omega_0}{2}}{2} \right]$$

$$\frac{2\omega - \omega_0}{2}$$

rect =  $\left[ \frac{\omega + \omega_0/2}{2\omega_0} \right]$

~~2021/11/11~~